

Strict Class 1 Reservation vs FCFS: Utilization, Service-Rate, and Waiting-Time Tradeoffs

Short preliminary decision memo for review

Executive Summary

- Bottom line: these 5-seed results are preliminary. Near the baseline, strict reservation does not beat FCFS on the utilization-first objective once the service-rate guardrail is checked.
- Baseline setting: $\lambda_1=\lambda_2=25$, total $\lambda=50$, $S=32$ slots/day, $H=14$ days, seeds 5101-5105.
- Study A objective is $U_{\text{util}} = \rho^{\text{booked}}$. At baseline it selects $Q=0$; strict-minus-FCFS booked utilization is $+0.000$; feasible=True.
- Study B objective is $U_{\text{wait_util}}$ with $w_1=1.5$, $c=0.05$, $\gamma=0.05$. At baseline it selects $Q=0$; strict-minus-FCFS utility is $+0.000$; feasible=True.
- The wait penalty does not change the selected baseline Q relative to Study A.
- Service-rate guardrails matter: cells marked with x in the heatmaps fail $\min(\text{served_rate_1}, \text{served_rate_2}) \geq 0.55$ and should not be interpreted as operational wins.
- Across the 12 demand/behavior scenarios, the two studies select the same Q in 4 cases. Differences mostly occur when wait reduction and Class 1 priority favor large Q but the service-rate guardrail flags infeasibility.
- Decision point: decide whether booked-slot utilization is the right primary capacity target, or whether completed visits and Class 2 access should receive stronger constraints before choosing Q .

1. What The Policy Changes

Pooled FCFS uses one common appointment pool. Strict Class 1 reservation protects Q slots per day for Class 1. Class 1 can use protected slots first and

then general slots. Class 2 can only use general slots. Unused protected slots remain empty.

The objective functions in this memo are after-the-fact scoring rules. They do not change the simulation policy.

2. Why Utilization Alone Is Not Enough

Booked utilization ρ^{booked} measures whether service-day appointment slots were booked. It includes no-shows because those slots were occupied on the schedule. Attended utilization ρ^{attended} measures completed visits. Served rate measures completed visits divided by arrivals.

A high booked-utilization value can still be unacceptable if one class has poor service. For that reason, all best-Q decisions are checked against $\min(\text{served_rate_1}, \text{served_rate_2}) \geq 0.55$.

3. Study A: Utilization-First With Guardrails

Main objective used to choose Q:

$$U_{\text{util}} = \rho^{\text{booked}} = \text{booked service-day slots} / \text{available measured slots}$$

The guardrail is not part of the score; it determines whether the selected Q is operationally acceptable.

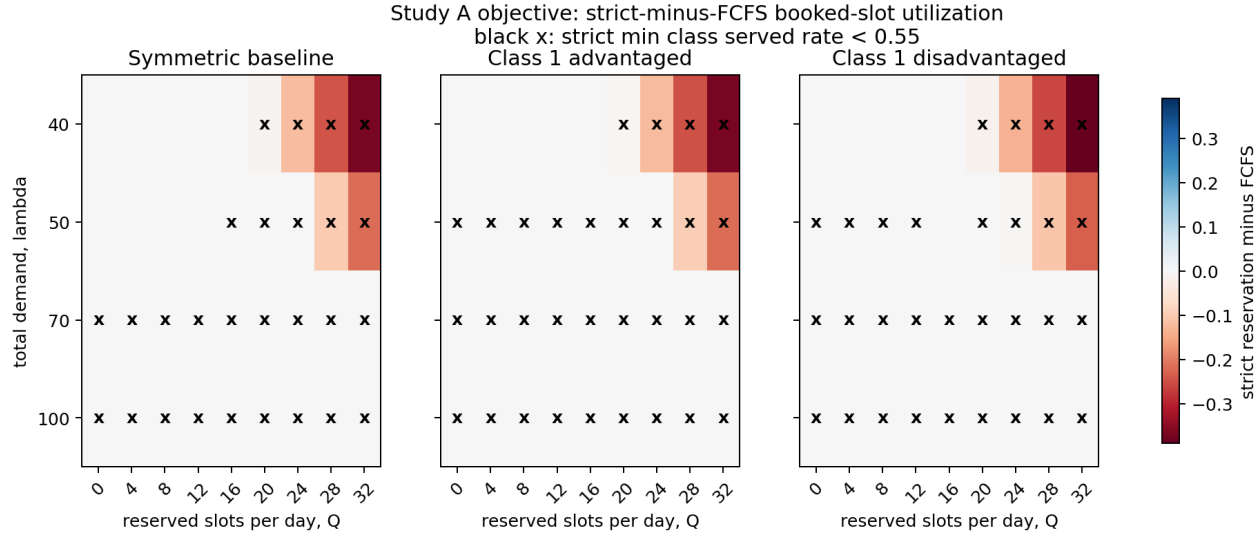


Figure 1. Study A objective: strict-minus-FCFS booked-slot utilization. Black x marks cells where the strict-reservation run violates the served-rate threshold.

4. Study B: Wait-Adjusted Utility With Guardrails

Main objective used to choose Q:

$U_wait_util = U_weighted_booked(w1, w2) - c \cdot Q/S - \gamma \tau_bar_w/H$,
with $w1=1.5$, $w2=1.0$, $c=0.05$, $\gamma=0.05$.

This score rewards booked capacity, gives Class 1 extra weight, penalizes protected capacity, and penalizes longer offered waits.

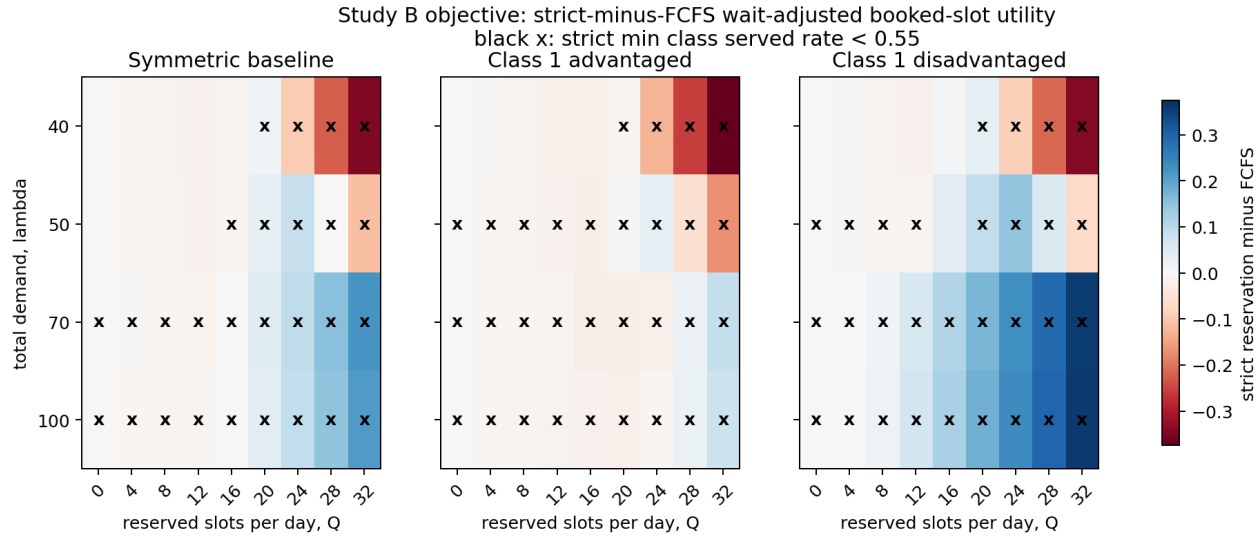


Figure 2. Study B objective: strict-minus-FCFS wait-adjusted booked-slot utility. Black x marks cells where the strict-reservation run violates the served-rate threshold.

5. Preliminary Results And Best-Q Comparison

Baseline policy comparison:

policy	Q	booked utilization	attended utilization	Class 1 utilization
Pooled FCFS	0	1.000	0.912	0.58
Study A/B selected Q=0 (same booking rule as FCFS)	0	1.000	0.912	0.58
Strict reservation, smallest positive Q=4	4	1.000	0.912	0.58

Best-Q comparison by study:

scenario	best Q under Study A	Study A feasible?	best Q under Study B	S
Symmetric baseline, lambda=40	0	yes	0	y
Symmetric baseline, lambda=50	0	yes	0	y
Symmetric baseline, lambda=70	0	no	32	r
Symmetric baseline, lambda=100	0	no	32	r
Class 1 advantaged, lambda=40	0	yes	0	y
Class 1 advantaged, lambda=50	0	no	24	r
Class 1 advantaged, lambda=70	0	no	32	r
Class 1 advantaged, lambda=100	0	no	32	r
Class 1 disadvantaged, lambda=40	0	yes	16	y
Class 1 disadvantaged, lambda=50	16	yes	16	y
Class 1 disadvantaged, lambda=70	0	no	32	r
Class 1 disadvantaged, lambda=100	0	no	32	r

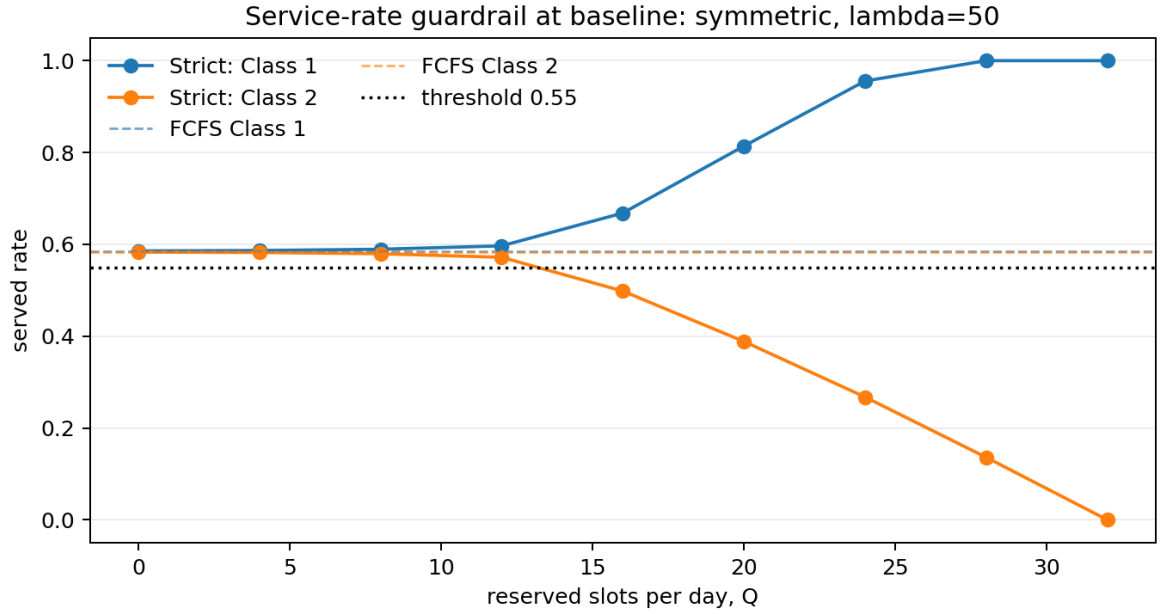


Figure 3. Service-rate diagnostic at the baseline scenario. Dashed lines are FCFS benchmarks; the dotted line is the 0.55 threshold.

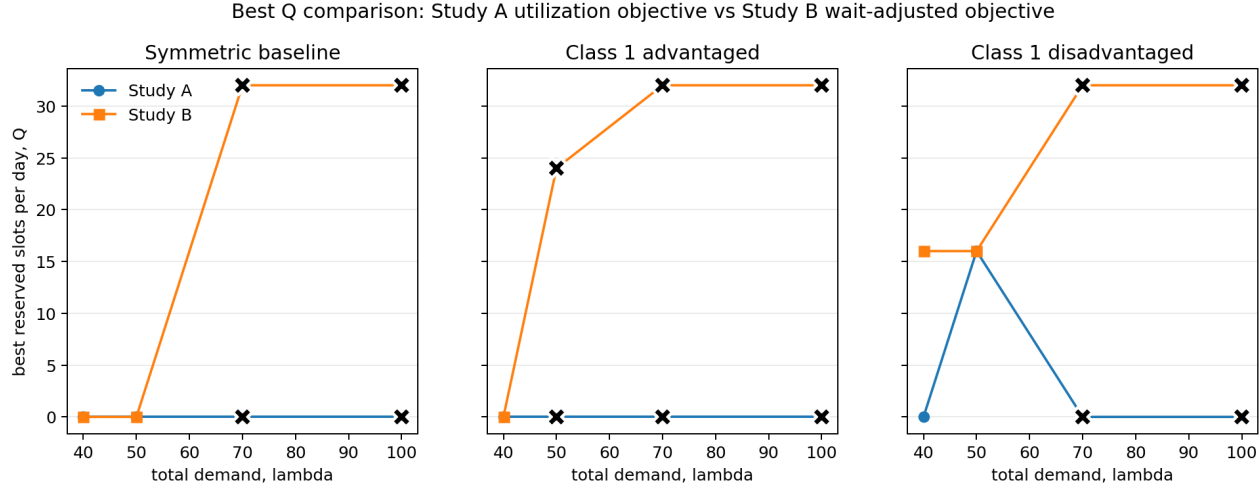


Figure 4. Best Q under each study. Black X markers with a white outline mark infeasible selected Q values.

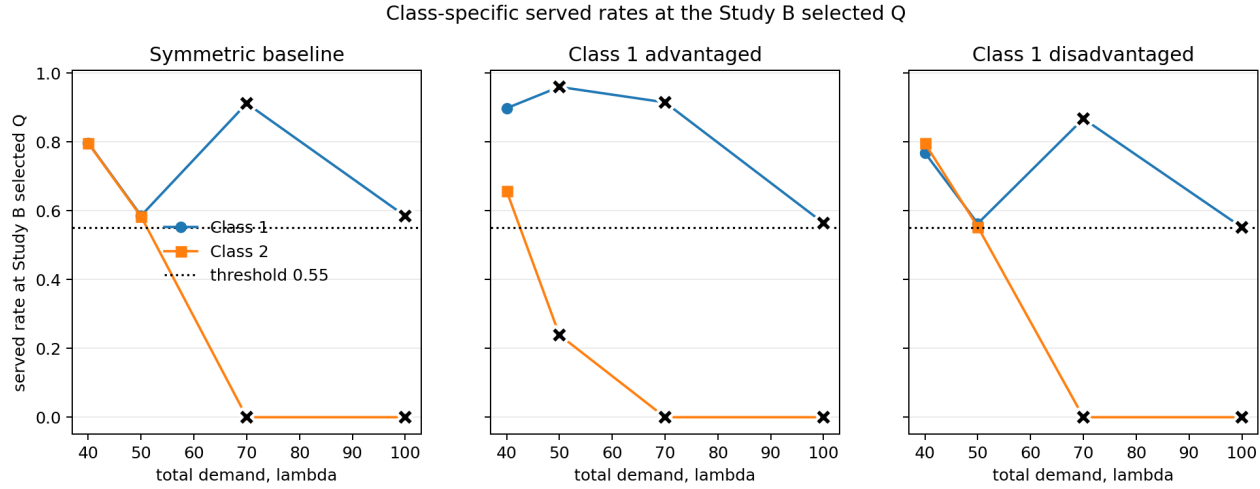


Figure 5. Class-specific served rates at the Study B selected Q. Class 1 and Class 2 are shown on the same axes; black X markers with a white outline mark infeasible selected Q values.

Summary comparison: the two studies agree in 4 of 12 scenarios. Study A is the more operationally defensible first-pass rule because it directly tests slot use and does not reward Class 1 access gains that come with large Class 2 service-rate losses. Study B is useful for studying priority and wait tradeoffs, but only after the service-rate guardrail is enforced.

6. Sensitivity To w1, c, gamma, And served_rate_min

study	w1	c	gamma	served_rate_min	best Q	strict beats FCFS?	binding const
B: wait-adjusted	1.500	0.000	0.000	0.550	12	yes	none
B: wait-adjusted	1.500	0.000	0.020	0.550	12	yes	none
B: wait-adjusted	1.500	0.000	0.050	0.550	12	yes	none
B: wait-adjusted	1.500	0.000	0.100	0.550	12	yes	none
B: wait-adjusted	1.500	0.050	0.000	0.550	0	no	objective does
B: wait-adjusted	1.500	0.050	0.020	0.550	0	no	objective does
B: wait-adjusted	1.500	0.050	0.050	0.550	0	no	objective does
B: wait-adjusted	1.500	0.050	0.100	0.550	0	no	objective does
B: wait-adjusted	1.500	0.100	0.000	0.550	0	no	objective does
B: wait-adjusted	1.500	0.100	0.020	0.550	0	no	objective does
B: wait-adjusted	1.500	0.100	0.050	0.550	0	no	objective does
B: wait-adjusted	1.500	0.100	0.100	0.550	0	no	objective does

Study B sensitivity at baseline: symmetric, lambda=50, w1=1.5

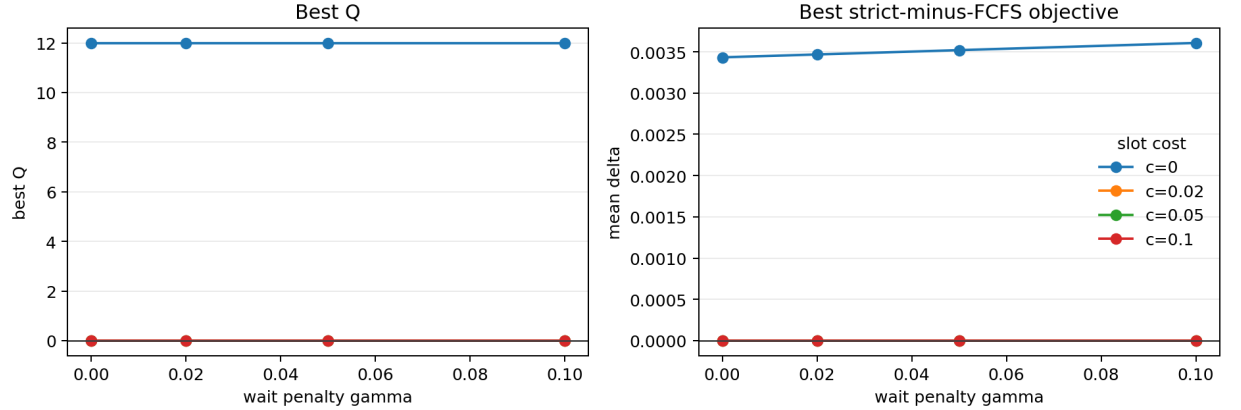


Figure 6. Study B sensitivity at the baseline scenario. The plotted objective is the best strict-minus-FCFS wait-adjusted utility for each gamma and slot-cost value.

7. Decision Points Before Finalizing

- Confirm whether the service-rate floor should be 0.55 or higher.
- Decide whether booked utilization is sufficient, or whether attended utilization should be a hard constraint because no-shows do not produce visits.

- Decide whether Class 2 service-rate loss should have its own explicit threshold.
- Rerun with more seeds before treating any Q recommendation as final.

Audit Notes

- Existing DOCX old-objective terms found: False.
- Old net-priority plot files found and not reused: True.
- booked_slot_utilization available in metrics layer/output: True.
- Class offered-delay totals available: True.
- Class service-rate guardrails computable: True.
- Each study uses exactly one main objective for best-Q selection.